

ITSimplera Institute
Summer Internship Program 2026



WEEK 01 ASSIGNMENT

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Implementation of a Basic LAN Topology in Cisco Packet Tracer

Introduction

Computer networking is the process of connecting two or more devices to enable communication and resource sharing. Networks are widely used in homes, educational institutions, businesses, and organizations to exchange information efficiently. Understanding networking fundamentals is essential for anyone pursuing a career in network administration or cybersecurity.

As part of the Week 01 internship assignment, a simple Local Area Network (LAN) was designed and implemented using Cisco Packet Tracer. The objective was to become familiar with the Packet Tracer interface, connect multiple devices using a network switch, assign static IP addresses, and verify communication between devices using the ping command. This hands-on activity provided practical experience with basic network configuration and troubleshooting.

Networking Concepts Learned

During this task, the following networking concepts were studied and implemented:

Computer Network

A computer network is a collection of interconnected devices that communicate with each other to share data and network resources.

Local Area Network (LAN)

A Local Area Network (LAN) connects devices within a limited geographical area such as a classroom, office, or home. LANs provide fast communication and efficient resource sharing.

IP Address

An IP address is a unique logical address assigned to every device connected to a network. It allows devices to identify and communicate with each other.

Subnet Mask

A subnet mask separates the network portion and host portion of an IP address. In this project, the subnet mask **255.255.255.0** was used.

Network Switch

A switch is a Layer 2 networking device that connects multiple devices within a LAN and forwards data based on MAC addresses. It improves communication efficiency by sending frames only to the intended destination.

Ping Command

The ping command is used to verify network connectivity between devices. Successful ping replies indicate that communication between hosts is functioning correctly.

Cisco Packet Tracer Interface Overview

Cisco Packet Tracer is a network simulation tool that allows users to design, configure, and troubleshoot virtual networks without physical networking equipment.

The main components of the interface explored during this assignment include:

- Workspace used to create and organize network topologies.
- Device Selection Panel containing routers, switches, end devices, and cables.
- Connection Tools used to connect devices with appropriate network cables.
- Logical Workspace for designing network topologies.
- Real-time Mode for immediate network simulation.
- Simulation Mode for observing packet flow and troubleshooting communication.

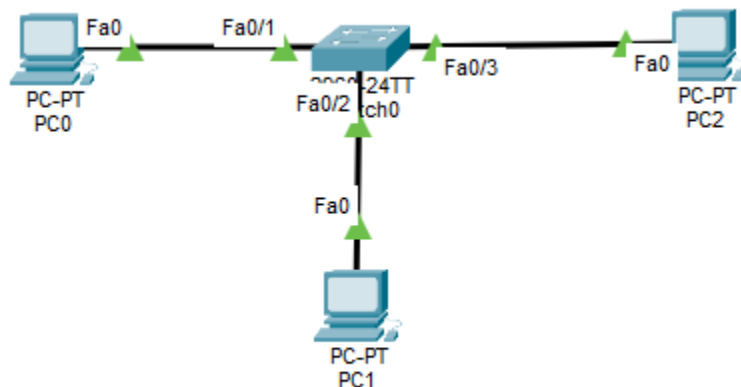
The software provides an easy-to-use environment for learning networking concepts through practical implementation.

LAN Topology Implementation

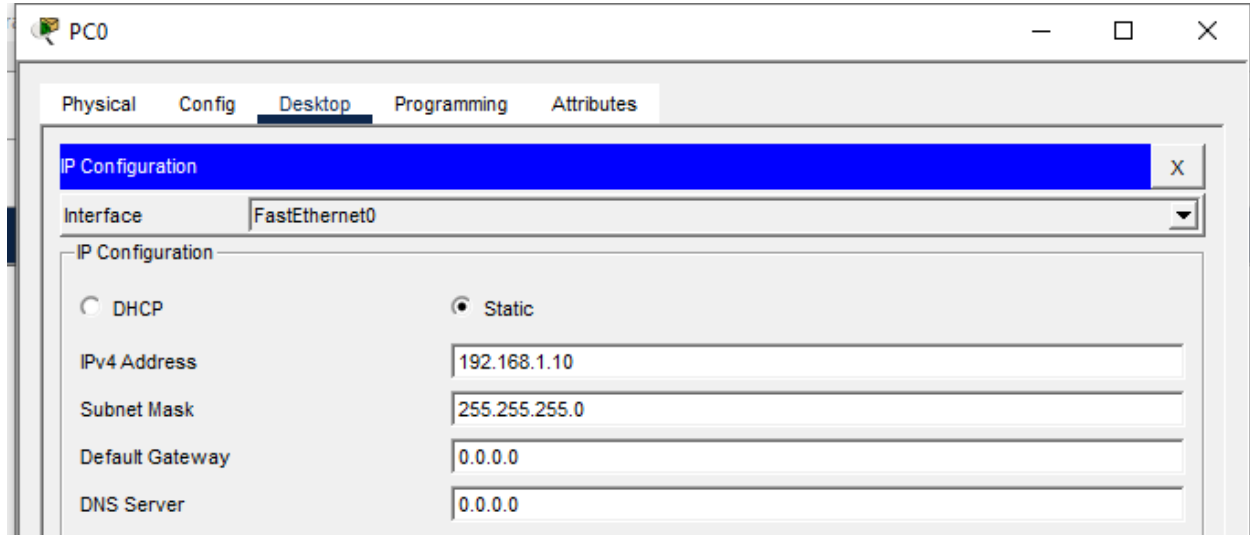
A simple LAN topology was created using one Cisco 2960 switch and three personal computers (PCs). Each computer was connected to the switch using Copper Straight-Through cables. The following static IP addresses were assigned:

Device	IP Address	Subnet Mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.11	255.255.255.0
PC2	192.168.1.12	255.255.255.0

Since all devices belong to the same subnet, no default gateway configuration was required.



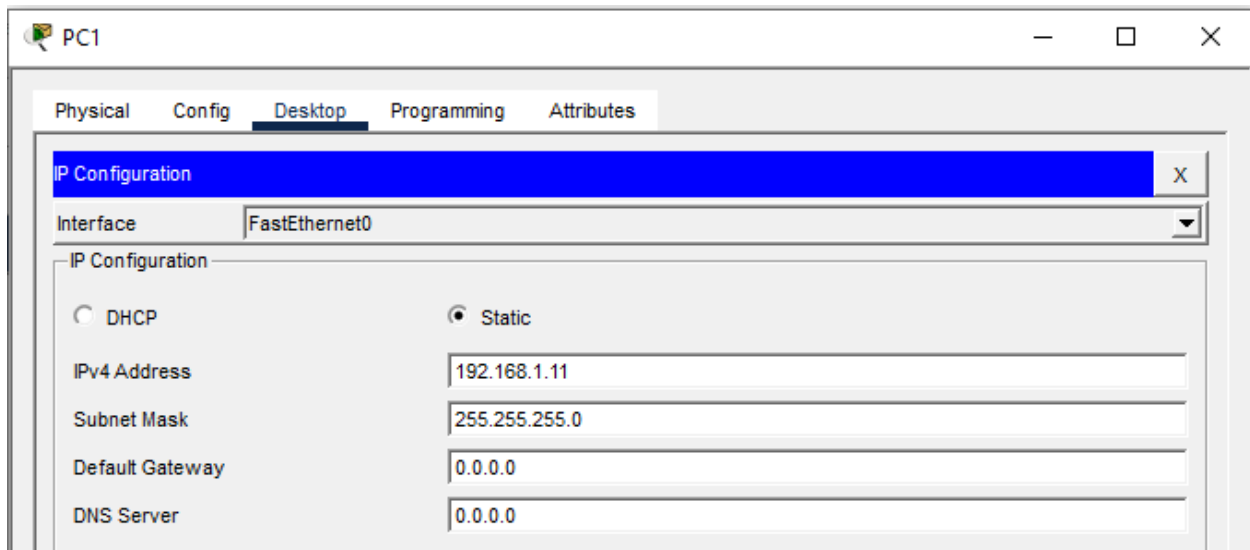
➤ For PC0



The screenshot shows the configuration window for PC0. The 'Desktop' tab is selected. The 'IP Configuration' section is expanded, showing the 'FastEthernet0' interface. The 'Static' radio button is selected, and the IP address is set to 192.168.1.10. The subnet mask is 255.255.255.0, and the default gateway and DNS server are both 0.0.0.0.

Interface	IP Configuration
FastEthernet0	☐ DHCP ☒ Static IPv4 Address: 192.168.1.10 Subnet Mask: 255.255.255.0 Default Gateway: 0.0.0.0 DNS Server: 0.0.0.0

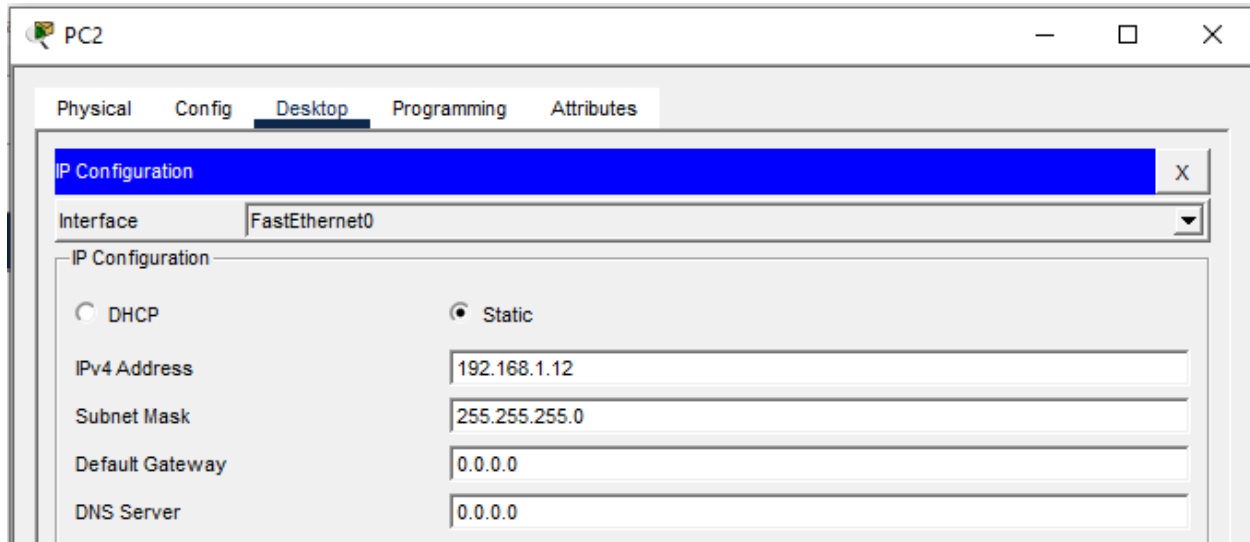
➤ For PC1



The screenshot shows the configuration window for PC1. The 'Desktop' tab is selected. The 'IP Configuration' section is expanded, showing the 'FastEthernet0' interface. The 'Static' radio button is selected, and the IP address is set to 192.168.1.11. The subnet mask is 255.255.255.0, and the default gateway and DNS server are both 0.0.0.0.

Interface	IP Configuration
FastEthernet0	☐ DHCP ☒ Static IPv4 Address: 192.168.1.11 Subnet Mask: 255.255.255.0 Default Gateway: 0.0.0.0 DNS Server: 0.0.0.0

- For PC2

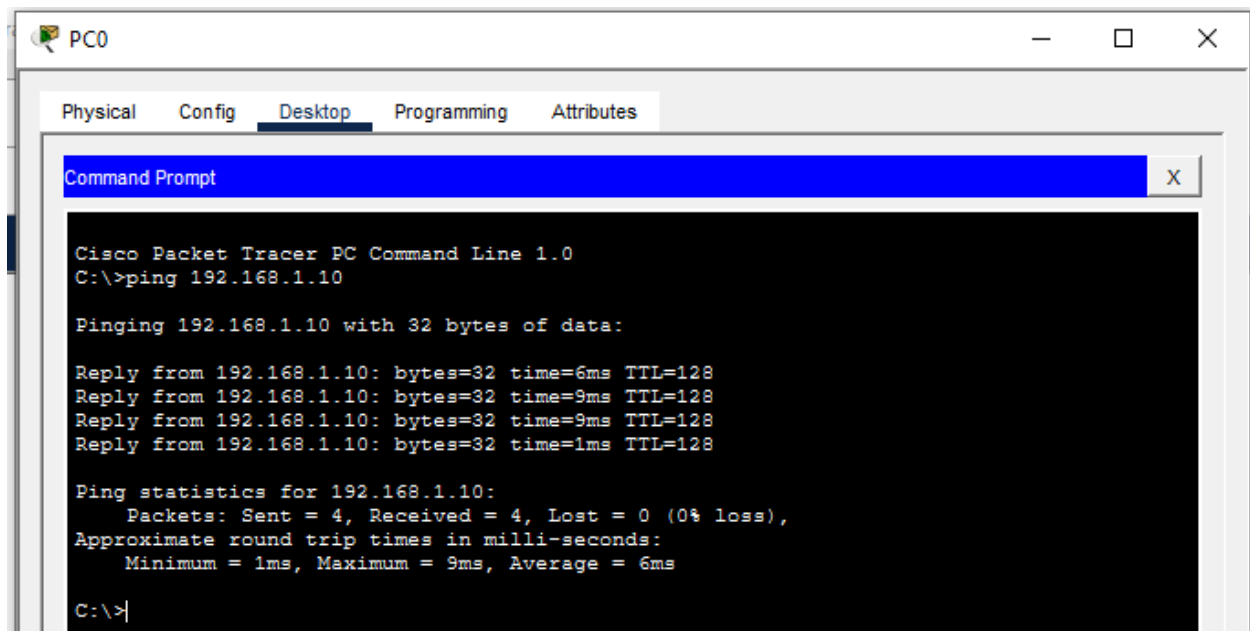


Connectivity Testing

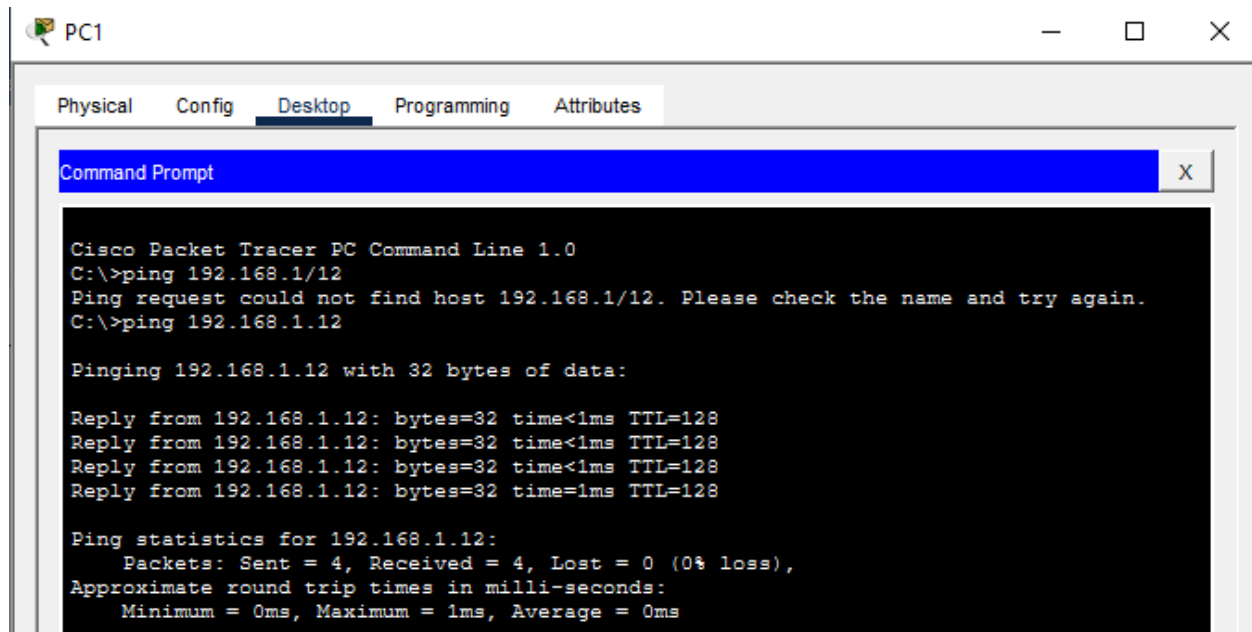
After assigning IP addresses, connectivity was verified using the ping command. Ping tests were performed from PC0 to PC1 and PC2.

The successful ping replies confirmed that all devices were correctly configured and capable of communicating within the Local Area Network.

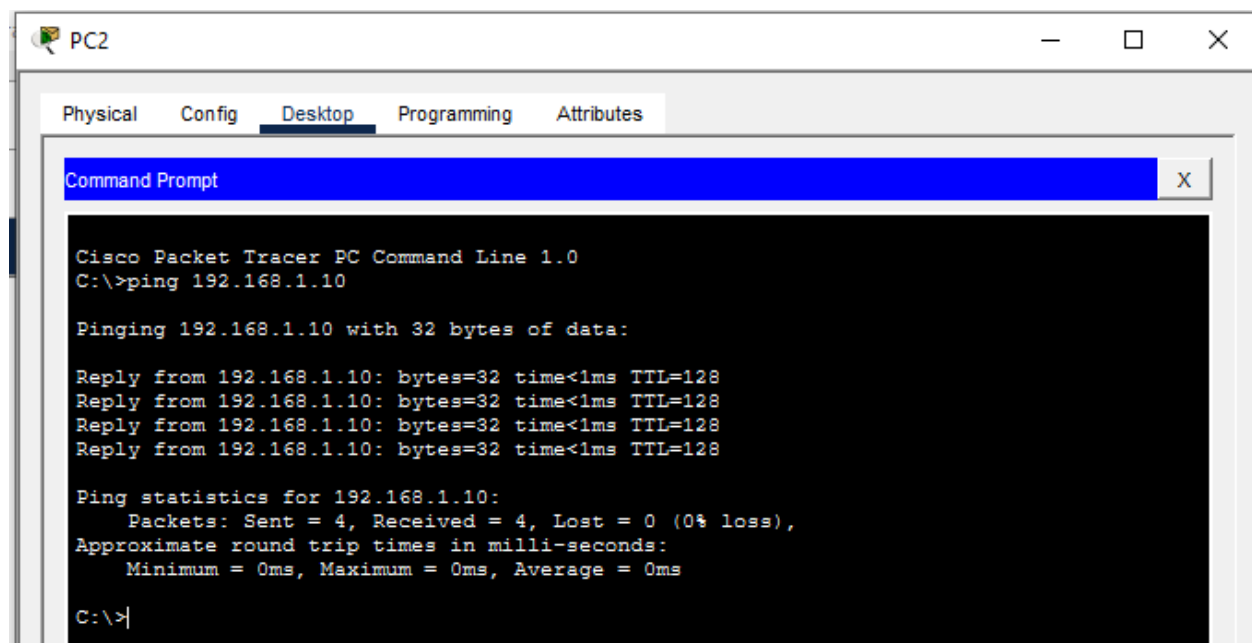
- Connectivity from PC0 to PC1



➤ Connectivity from PC1 to PC2



➤ Connectivity from PC2 to PC0



Conclusion

This practical assignment provided valuable hands-on experience with Cisco Packet Tracer and reinforced the fundamental concepts of computer networking. A simple LAN topology was successfully created using one switch and three computers. Static IP addressing was configured correctly, and connectivity was verified through successful ping tests. Overall, the activity enhanced practical networking, troubleshooting, and documentation skills while providing a strong foundation for more advanced networking concepts in future internship tasks.

Implementation of a Basic LAN Topology in GNS3

Introduction

GNS3 (Graphical Network Simulator-3) is a powerful network simulation platform used by students, network engineers, and IT professionals to design, configure, and test network topologies in a virtual environment. It enables users to simulate real networking devices and gain practical experience without requiring physical hardware.

As part of the Week 01 internship assignment, a simple Local Area Network (LAN) was designed and implemented using GNS3. The objective was to become familiar with the GNS3 interface, create a basic network topology, configure static IP addresses on virtual PCs, and verify network connectivity using the ping command. This practical activity strengthened fundamental networking concepts and provided hands-on experience with network simulation.

Networking Concepts Learned

During this practical exercise, the following networking concepts were explored:

Computer Network

A computer network is a collection of interconnected devices that exchange information and share resources through communication protocols.

Local Area Network (LAN)

A Local Area Network (LAN) connects computers and other network devices within a limited geographical area such as a home, office, or educational institution.

IP Address

An IP address uniquely identifies each device connected to a network and allows communication between network hosts.

Subnet Mask

A subnet mask defines the network portion and host portion of an IP address. The subnet mask used in this project was **255.255.255.0 (/24)**.

Ethernet Switch

An Ethernet switch connects multiple devices within the same LAN and forwards data frames based on MAC addresses, enabling efficient communication between connected hosts.

Ping Command

The ping command is a network diagnostic utility used to verify communication between devices by sending Internet Control Message Protocol (ICMP) Echo Request packets.

GNS3 Interface Overview

GNS3 is an advanced network simulation and emulation platform that allows users to build realistic network environments using virtual networking devices.

The primary components of the GNS3 interface include:

- **Project Workspace** for designing and managing network topologies.
- **Device Toolbar** containing switches, routers, virtual PCs (VPCS), and other appliances.
- **Node List** displaying available networking devices.
- **Console Access** for configuring and managing virtual devices.
- **Start, Stop, and Reload Controls** for managing the status of network devices.
- **Packet Capture Feature** for analyzing network traffic using Wireshark.

The interface provides a realistic environment for practicing network configuration, testing, and troubleshooting.

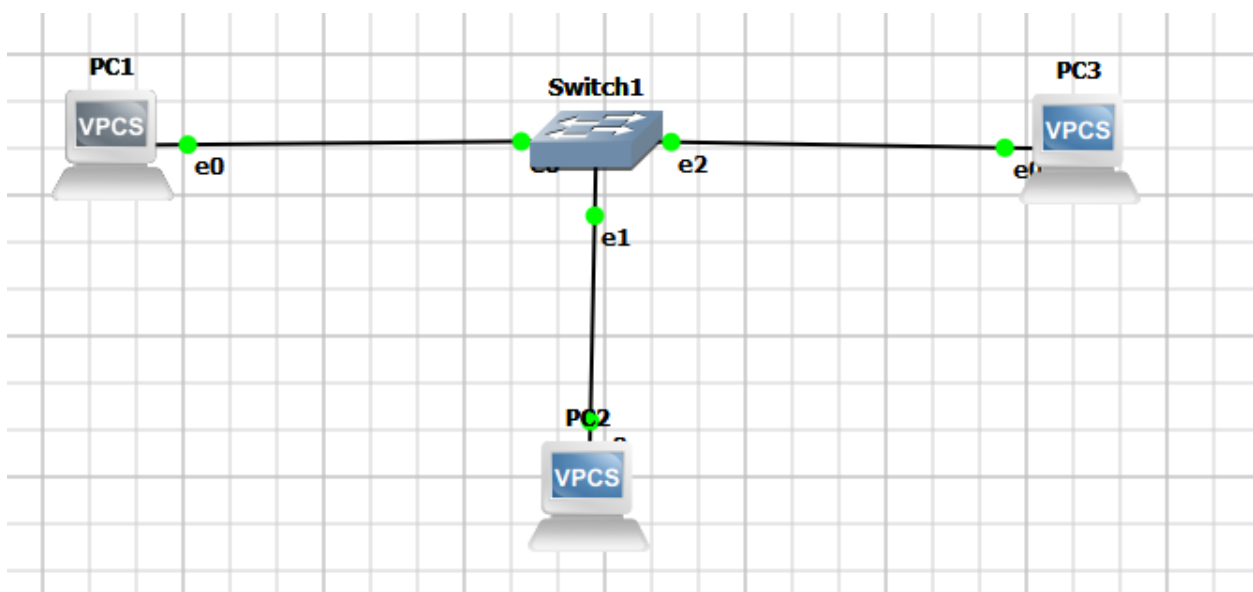
LAN Topology Implementation

A basic LAN topology was created using one Ethernet Switch and three Virtual PCs (VPCS). Each virtual PC was connected to the switch using Ethernet links.

The following static IP addresses were assigned:

Device	IP Address	Subnet Mask
VPCS1	192.168.1.10	255.255.255.0
VPCS2	192.168.1.11	255.255.255.0
VPCS3	192.168.1.12	255.255.255.0

Since all devices were configured within the same subnet, no default gateway was required for communication.



➤ For pc1

```
PC1> ip 192.168.1.10 255.255.255.0 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1
PC1> 
```

➤ For pc2

```
PC2> ip 192.168.1.11
Checking for duplicate address...
PC1 : 192.168.1.11 255.255.255.0
```

➤ For pc3

```
Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC3> ip 192.168.1.12 255.255.255.0 192.168.1.0
Invalid gateway address

PC3> ip 192.168.1.12 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.12 255.255.255.0
PC3> 
```

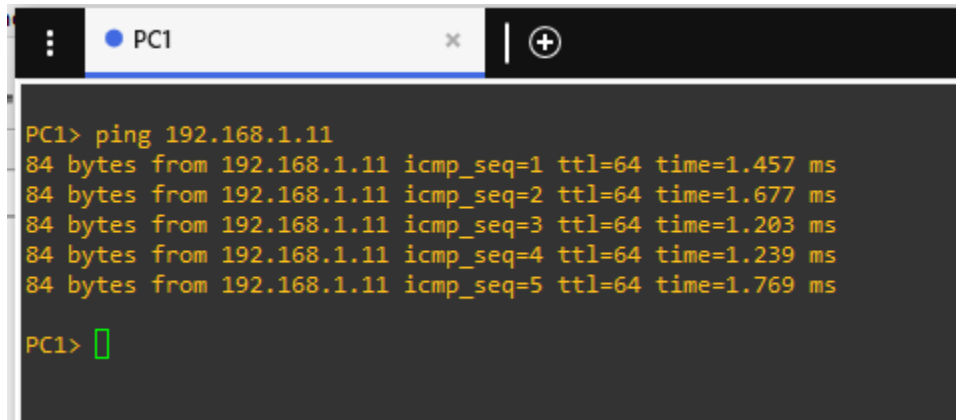
Connectivity Testing

After configuring the IP addresses, connectivity between all virtual PCs was tested using the ping command.

Ping tests were successfully performed from VPCS1 to VPCS2 and VPCS3.

The successful ping replies confirmed that the LAN topology was correctly configured and that all devices were able to communicate within the network.

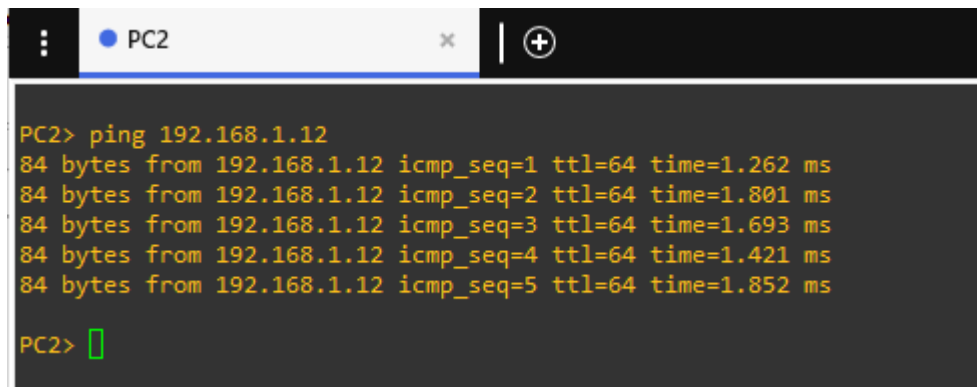
➤ Connectivity from pc1 to pc2



```
PC1> ping 192.168.1.11
84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=1.457 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=1.677 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=1.203 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=1.239 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=1.769 ms

PC1> 
```

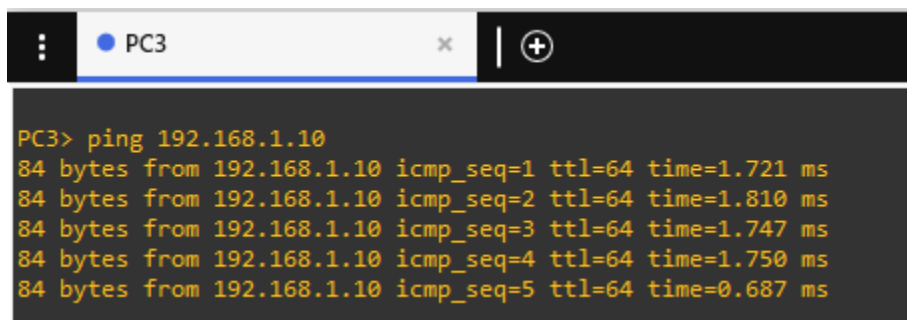
➤ Connectivity from pc2 to pc2



```
PC2> ping 192.168.1.12
84 bytes from 192.168.1.12 icmp_seq=1 ttl=64 time=1.262 ms
84 bytes from 192.168.1.12 icmp_seq=2 ttl=64 time=1.801 ms
84 bytes from 192.168.1.12 icmp_seq=3 ttl=64 time=1.693 ms
84 bytes from 192.168.1.12 icmp_seq=4 ttl=64 time=1.421 ms
84 bytes from 192.168.1.12 icmp_seq=5 ttl=64 time=1.852 ms

PC2> 
```

➤ Connectivity from pc3 to pc1



```
PC3> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=64 time=1.721 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=64 time=1.810 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=64 time=1.747 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=64 time=1.750 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=64 time=0.687 ms
```

Conclusion

This practical assignment provided valuable hands-on experience with GNS3 and reinforced fundamental networking concepts through practical implementation. A simple Local Area Network (LAN) was successfully designed using an Ethernet Switch and Virtual PCs. Static IP addresses were configured correctly, and communication between devices was verified using the ping command. The activity improved practical networking, troubleshooting, and documentation skills while building confidence in using GNS3 for network simulation and future networking projects.